Guia 6 Ej 1

Movimiento de un Fluido con Componentes de Velocidad u y derivados de un Potencial ϕ

$U = \frac{\partial \phi}{\partial x}$ $\quad V = \frac{\partial \phi}{\partial y}$ $\quad w = 0$

a) $\phi = \frac{1}{4\pi} \operatorname{Log}(x^2+y^2) = \frac{1}{2\pi} \operatorname{Log} r$ $\quad r^2 = x^2+y^2$

b) $\phi = x$

c) $\phi = A r^n \cos m\theta$ $\quad \theta = \tan^{-1} \frac{y}{x}$

d) $\phi = \frac{\cos\theta}{r}$

a) $U = \frac{\partial \phi}{\partial x} \rightarrow U = \frac{\partial}{\partial x}\left(\frac{1}{4\pi} \log(x^2+y^2)\right)$

$$\left[\log u \quad u = x^2+y^2 \quad \frac{1}{u} \cdot 2x = \frac{1}{x^2+y^2} \cdot 2x \right.$$

$U = \frac{1}{4\pi} \cdot \frac{\partial}{\partial x}\left(\log(x^2+y^2)\right)$

$U = \frac{1}{4\pi} \cdot \frac{1}{(x^2+y^2)} \cdot 2x$

$$\left[U = \frac{x}{2\pi(x^2+y^2)} \right]$$

$V = \frac{\partial \phi}{\partial y} = \frac{\partial}{\partial y}\left(\frac{1}{4\pi} \operatorname{Log}(x^2+y^2)\right)$

$V = \frac{1}{4\pi} \cdot \frac{1}{(x^2+y^2)} 2y$

$$\left[V = \frac{y}{2\pi(x^2+y^2)} \right]$$

b)

$$U = \frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x}(x)$$

$$U = 1$$

$$V = \frac{\partial \phi}{\partial y} \rightarrow \frac{\partial}{\partial y}(x)$$

$$V = 0$$

c) $\phi = A r^n \cos n\theta$

$\phi(r,\phi) = \phi(x(r,\phi), y(r,\phi))$ $\quad x = r\cos\theta \quad y = r \,\text{sen}\,\theta$

$$\frac{\partial \phi(r,\theta)}{\partial r} = \frac{\partial \phi(x,y)}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial \phi(x,y)}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\frac{\partial \phi}{\partial r}(r,\phi) = U \cdot \cos\theta + V \cdot \text{sen}\,\theta$$

$$\frac{\partial \phi}{\partial r}(r,\theta) = Ur$$

$$\frac{\partial \phi}{\partial \phi}(r,\phi) = \frac{\partial \phi(x,y)}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial \phi(x,y)}{\partial y} \cdot \frac{\partial y}{\partial \theta}$$

$$= -U \cdot r\,\text{sen}\,\theta + V \cdot r \cdot \cos\theta$$

$$= r \cdot (V \cdot \cos\theta - U \cdot \text{sen}\,\theta$$

$$= r \cdot U\theta$$

Entonces $\quad U\theta = \frac{1}{r} \cdot \frac{\partial \phi(r,\theta)}{\partial \phi}$

Para $\phi = A \cdot r^n \cdot \cos(n\theta)$

$$Ur = \frac{\partial \phi}{\partial r} = A \cdot n \cdot r^{n-1} \cos(n\theta$$

$$U\theta = \frac{1}{r} \cdot \frac{\partial \phi}{\partial \theta} = \frac{1}{r} \cdot A \cdot r^n \cdot (-\text{sen}(n\theta))n = A \cdot r^n \cdot r^{n-1} \cdot n \cdot \text{sen}(n\theta)$$

$$U_\theta = -A n \cdot r^{n-1} \cdot \text{sen}(n\theta)$$

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d) $\phi = \frac{\cos\phi}{r} = \cos\theta \cdot r^{-1}$

$U_r = \frac{\partial\phi}{\partial r} \cdot \cos\theta \cdot (-1) \cdot r^{-2} = \frac{-\cos\theta}{r^2}$

$U_\theta = \frac{1}{r} \cdot \frac{\partial\phi}{\partial\theta} = \frac{1}{r} \cdot \frac{1}{r} \cdot -\sin\theta = \frac{-\sin\theta}{r^2}$

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$$u = \frac{\partial \psi}{\partial y} \qquad V = -\frac{\partial \psi}{\partial x}$$

$$\psi(r,\theta) = \psi(x(r,\theta), y(r,\theta))$$

$$\frac{d\psi(r,\theta)}{d\theta} = \left[\frac{\partial \psi(x,y)}{\partial x} \cdot \frac{dx}{d\theta} + \frac{\partial \psi(x,y)}{\partial y} \cdot \frac{dy}{d\theta}\right]$$

$$\frac{d\psi(r,\theta)}{d\theta} = \left[-(V) \cdot (-r \operatorname{sen}\theta) + (u) \cdot (r\cos\theta)\right]$$

$$\frac{d\psi(r,\theta)}{d\theta} = r \cdot V \cdot \operatorname{sen}\theta + r \cdot u \cdot \cos\theta$$

$$= r \cdot (V \cdot \operatorname{sen}\theta + u \cdot \cos\theta)$$

$$\frac{d\psi(r,\theta)}{d\theta} = r \cdot u_r$$

$$\left[U_r = \frac{1}{r} \frac{d\psi(r,\theta)}{d\theta} \right]$$

$$\frac{d\psi(r,\theta)}{dr} = \frac{d\psi(x,y)}{dx} \cdot \frac{dx}{dr} + \frac{d\psi(x,y)}{dy} \cdot \frac{dy}{dr}$$

$$\frac{d\psi(r,\theta)}{dr} = -V \cdot \cos\theta + u \cdot \operatorname{sen}\theta$$

$$= -(V \cdot \cos\theta - u \cdot \operatorname{sen}\theta$$

$$\frac{d\psi(r,\theta)}{dr} = -u_\theta$$

$$U_\theta = -\frac{\partial \psi(r,\theta)}{dr}$$

a) $\psi = C \cdot \theta$

$$U_r = \frac{1}{r} \cdot C = \frac{C}{r}$$

$$U_\theta = 0$$

b) $\psi = y$

$$u = 1$$

$$V = 0$$

c) $\Psi = A \cdot r^m \sin(m\theta)$

$$U_r = \frac{1}{r} \cdot \frac{\partial \Psi}{\partial \theta} = \frac{1}{r} \cdot A \cdot r^m \cdot \cos(m\theta) \cdot m$$

$$U_r = Am \cdot r^{m-1} \cdot \cos(m\theta)$$

$$U_\theta = \frac{-\partial \Psi}{\partial r} = -A \cdot m \cdot r^{m-1} \cdot \sin(m \cdot \theta)$$

d) $\Psi = -\frac{\sin\theta}{r} = -\sin\theta \cdot r^{-1}$

$$U_r = \frac{1}{r} \cdot \left(-\frac{1}{r} \cdot \cos\theta\right) \quad U_r = -\frac{\cos\theta}{r^2}$$

$$U_\theta = -\frac{\partial \Psi}{\partial r} = -(\sin\theta) \cdot (-1) \cdot r^{-2} = -\frac{\sin\theta}{r^2}$$

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Vorticidad: $\Omega = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) = \frac{1}{2}\left(\frac{\partial u_\theta}{\partial r} - \frac{1}{r}\frac{\partial u_r}{\partial \theta} + \frac{u_\theta}{r}\right)$

Mostrar que la vorticidad desaparece en cada caso

a) $\begin{bmatrix} v = \frac{1}{2\pi} \cdot \frac{y}{(x^2+y^2)} \\ u = \frac{1}{2\pi} \cdot \frac{x}{(x^2+y^2)} \end{bmatrix}$ $\quad \frac{dv}{dx} = \frac{1}{2\pi} \cdot y \cdot (-1) \cdot (x^2+y^2)^{-2} \cdot 2x$

$$\frac{dv}{dx} = -\frac{1}{\pi} \cdot \frac{xy}{(x^2+y^2)^2} \quad (1)$$

$$\frac{\partial u}{\partial y} = \frac{1}{2\pi} \cdot x \cdot (-1) \cdot (x^2+y^2)^{-2} \cdot 2y$$

$$\frac{\partial u}{\partial y} = -\frac{1}{\pi} \cdot \frac{x \cdot y}{(x^2+y^2)^2} \quad (2)$$

$$\Omega = \frac{1}{2} \cdot \left(-\frac{1}{\pi} \cdot \frac{xy}{(x+y)^2} - \left(-\frac{1}{\pi}\frac{xy}{(x+2)^2}\right)\right) = \frac{1}{2} \cdot 0 = 0 \checkmark$$

b) $v = 0 \quad \frac{\partial v}{\partial x} = 0 \quad \Omega = \frac{1}{2} \cdot (0-0) = 0 \checkmark$

$u = 1 \quad \frac{\partial u}{\partial y} = 0$

c) $u_\theta = -A \cdot m \cdot r^{m-1} \cdot \sin(m\theta)$

$$\frac{\partial u_\theta}{\partial r} = -m(m-1) \cdot A \cdot r^{m-2} \cdot \sin(m\theta)$$

$$u_r = A \cdot m \cdot r^{m-1} \cdot \cos(m\theta)$$

$$\frac{\partial u_r}{\partial \theta} = -m^2 \cdot A \cdot r^{m-1} \cdot \sin(m\theta)$$

$$-\frac{1}{r} \cdot \frac{\partial u_r}{\partial \theta} = m^2 \cdot A \cdot r^{m-2} \cdot \sin(m\theta)$$

$$\frac{u_\theta}{r} = -m \cdot A \cdot r^{m-2} \cdot \sin(m\theta)$$

$$\Omega = \frac{1}{2}\cdot\left(-m.(m-1).A.r^{m-2}.\sin(m\theta) + m^2 \cdot A.r^{m-2}.\sin(m\theta) - m.A.r^{m-2}.\sin(m\theta)\right)$$

$$\Omega = \frac{1}{2}\left((-m(m-1)+m^2-m)\cdot A.r^{m-2}\cdot\sin(m\theta)\right)$$

$$\Omega = \frac{1}{2}\left(\underbrace{(-m^2+m+m^2-m)}_{0}\cdot A.r^{m-2}\cdot\sin(m\theta)\right)$$

$$\Omega = \frac{1}{2}\cdot 0 = 0 \checkmark$$

d)

$$U_\theta = \frac{-\sin\theta}{r^2} \qquad U_r = \frac{-\cos\theta}{r^2} \qquad -\frac{1}{r}\cdot\frac{\partial U_r}{\partial\theta} = \frac{-\sin\theta}{r^3}$$

$$\frac{\partial U_\theta}{\partial r} = \frac{2\cdot\sin\theta}{r^3} \qquad \frac{\partial U_r}{\partial\theta} = -\frac{1}{r^2}\cdot(-\sin\theta) \qquad \frac{U_\theta}{r} = -\frac{\sin\theta}{r^3}$$

$$\Omega = \frac{1}{2}\cdot\left(\frac{2\sin\theta}{r^3} - \frac{\sin\theta}{r^3} - \frac{\sin\theta}{r^3}\right)$$

$$\Omega = \frac{1}{2}\cdot\left(\frac{2\sin\theta}{r^3} - \frac{2\sin\theta}{r^3}\right) = \Omega = \frac{1}{2}\cdot(0) = 0 \checkmark$$

b) Expresiones para el Tensor Tasa de deformación

Tensor Tasa deformación: $V_{ij} = V_{ji}$ $\qquad V_{ij} = \frac{1}{2}\left(\frac{\partial V_j}{\partial x_i} + \frac{\partial V_i}{\partial x_j}\right) = \frac{1}{2}\left(\frac{\partial V}{\partial x} + \frac{\partial U}{\partial y}\right)$

En coordenadas polares

$$V_{rr} = \frac{\partial U_r}{\partial r} \qquad V_{\theta\theta} = \frac{U_r}{r} + \frac{1}{r}\cdot\frac{\partial U_\theta}{\partial\theta} \qquad V_{r\theta} = \left(\frac{1}{2}\cdot\left(\frac{\partial U_\theta}{\partial r} + \frac{1}{r}\cdot\frac{\partial U_r}{\partial\theta} - \frac{U_\theta}{r}\right)\right)$$

a) $\frac{\partial V}{\partial x} = \frac{1}{2\pi}\left(\frac{x^2+y^2-x}{(x^2+y^2)^2}\right)$

$$\frac{\partial U}{\partial y} = \frac{1}{2\pi}\cdot\left(\frac{x^2+y^2-y}{(x^2+y^2)^2}\right)$$

$$\left[V_{11} = \frac{\partial V}{\partial x} = \frac{1}{2\pi}\cdot\left(\frac{x^2+y^2-x}{(x^2+y^2)^2}\right)\right] \qquad V_{12} = \frac{1}{2}\left(-\frac{1}{\pi}\cdot\frac{x.y}{(x^2+y^2)^2} - \frac{1}{\pi}\cdot\frac{x.y}{(x^2+y^2)^2}\right)$$

$$\left[V_{22} = \frac{\partial U}{\partial y} = \frac{1}{2\pi}\cdot\left(\frac{x^2+y^2-y}{(x^2+y^2)^2}\right)\right] \qquad \left[V_{12} = -\frac{1}{\pi}\cdot\frac{x.y}{(x^2+y^2)^2}\right]$$

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b) $U = 1$ $\quad V = 0$

$$\frac{\partial U}{\partial x} = 0 \quad \frac{\partial V}{\partial y} = 0 \quad \frac{\partial V}{\partial x} = 0 \quad \frac{\partial U}{\partial y} = 0$$

$$V_{11} = \frac{\partial U}{\partial x} = 0 \qquad V_{12} = \frac{1}{2} \cdot (0 + 0) = 0$$

$$V_{22} = \frac{\partial V}{\partial y} = 0$$

c) $\frac{\partial U_r}{\partial r} = m \cdot (m-1) \cdot A \cdot r^{m-2} \cdot \cos(m\theta)$

$$\frac{\partial U_\theta}{\partial \theta} = -m^2 \cdot A \cdot r^{m-1} \cdot \cos(m\theta)$$

$$\frac{\partial U_\theta}{\partial r} = -m \cdot (m-1) \cdot A \cdot r^{m-2} \cdot \sin(m\theta)$$

$$\frac{\partial U_r}{\partial \theta} = -m^2 \cdot A \cdot r^{m-1} \cdot \sin(m\theta)$$

$$V_{rr} = \frac{\partial U_r}{\partial r} = m \cdot (m-1) \cdot A \cdot r^{m-2} \cdot \cos(m\theta)$$

$$V_{\theta\theta} = \frac{U_r}{r} + \frac{1}{r} \cdot \frac{\partial U_\theta}{\partial \theta} = \frac{m \cdot A \cdot r^{m-1} \cdot \cos(m\theta)}{r} + \frac{1}{r} \cdot (-m^2 \cdot A \cdot r^{m-1} \cdot \cos(m\theta))$$

$$V_{\theta\theta} = m \cdot (1-m) \cdot A \cdot r^{m-2} \cdot \cos(m\theta)$$

$$V_{r\theta} = \frac{1}{2}\left(\frac{\partial U_\theta}{\partial r} + \frac{1}{r} \cdot \frac{\partial U_r}{\partial \theta} - \frac{U_\theta}{r}\right)$$

$$V_{r\theta} = m \cdot (-m+1) \cdot A \cdot r^{m-2} \cdot \sin(m\theta)$$

d) $\frac{\partial U_r}{\partial r} = \frac{3 \cdot \cos\theta}{r^3}$ $\frac{\partial U_\theta}{\partial \theta} = -\frac{\cos\theta}{r^2}$ $\frac{\partial U_\theta}{\partial r} = \frac{2 \operatorname{sen}\theta}{r^3}$ $\frac{\partial U_r}{\partial \theta} = \frac{\operatorname{sen}\theta}{r^2}$

$$V_{rr} = \frac{\partial U_r}{\partial r} = \frac{2\cos\theta}{r^3}$$

$$V_{\theta\theta} = \frac{U_r}{r} + \frac{1}{r} \cdot \frac{\partial U_\theta}{\partial \theta}$$

$$V_{\theta\theta} = \frac{-2\cos\theta}{r^3}$$

$$V_{r\theta} = \frac{1}{2}\left(\frac{\partial U_\theta}{\partial r} + \frac{1}{r} \cdot \frac{\partial U_r}{\partial \theta} - \frac{U_\theta}{r}\right)$$

$$V_{r\theta} = \frac{2 \operatorname{sen}\theta}{r^3}$$

C) Verificar si los flujos resultan incompresibles

Realizado en Octave

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$U_1 = ax^2 + bxy + c \qquad V = by^2 + cx + mz \qquad W = mz^3$

$e_{ij} = \frac{1}{2}\left[\frac{\partial U_j}{\partial x_i} + \frac{\partial U_i}{\partial x_j}\right]$

$U_1 = a x_1^2 + b x_1 x_2 + c$
$U_2 = b x_2^2 + c x_1 + m x_3$
$U_3 = m x_3^3$

$\varepsilon_{11} = \frac{1}{2}\left[\frac{\partial U_1}{\partial x_1} + \frac{\partial U_1}{\partial x_1}\right] = \frac{1}{2}\left[2ax_1 + bx_2 + 2ax_1 + bx_2\right] = ax_1 + bx_2$

$\varepsilon_{12} = \frac{1}{2}\left[\frac{\partial U_2}{\partial x_1} + \frac{\partial U_1}{\partial x_2}\right] = \frac{1}{2}\left[c + bx_1\right]$

$\varepsilon_{13} = \frac{1}{2}\left[\frac{\partial U_3}{\partial x_1} + \frac{\partial U_1}{\partial x_3}\right] = \frac{1}{2}\left[0 + 0\right] = 0$

$\varepsilon_{21} = \frac{1}{2}\left[\frac{\partial U_1}{\partial x_2} + \frac{\partial U_2}{\partial x_1}\right] = \frac{1}{2}\left[bx_1 + c\right]$

$\varepsilon_{22} = \frac{1}{2}\left[\frac{\partial U_2}{\partial x_2} + \frac{\partial U_2}{\partial x_2}\right] = \frac{1}{2}\left[2bx_2 + 2bx_2\right] = \frac{1}{2} \cdot 2(bx_2 + bx_2) = 2bx_2$

$\varepsilon_{23} = \frac{1}{2}\left[\frac{\partial U_3}{\partial x_2} + \frac{\partial U_2}{\partial x_3}\right] = \frac{1}{2} \cdot \left[0 + m\right] = \frac{1}{2}m$

$\varepsilon_{31} = \frac{1}{2}\left[\frac{\partial U_1}{\partial x_3} + \frac{\partial U_3}{\partial x_1}\right] = \frac{1}{2}\left[0 + 0\right] = 0$

$\varepsilon_{32} = \frac{1}{2}\left[\frac{\partial U_2}{\partial x_3} + \frac{\partial U_3}{\partial x_2}\right] = \frac{1}{2}\left[m + 0\right] = \frac{1}{2}m$

$\varepsilon_{33} = \frac{1}{2}\left[\frac{\partial U_3}{\partial x_3} + \frac{\partial U_3}{\partial x_3}\right] = \frac{1}{2}\left[3mx_3^2 + 3mx_3^2\right] = \frac{1}{2} \cdot 3(mx_3^2 + mx_3^2) = \frac{3}{2} \cdot 2(mx_3^2)$

$\varepsilon_{33} = 3mx_3^2$

$\frac{\partial^2 e_{11}}{\partial x_2 \partial x_3} = 0 \qquad \frac{\partial}{\partial x_1}\left(-\frac{\partial e_{23}}{\partial x_1} + \frac{\partial e_{32}}{\partial x_2} + \frac{\partial e_{12}}{\partial x_3}\right)$

$\frac{\partial}{\partial x_1}\left(-0 + 0 + 0\right) = 0$ ✓

$[0 = 0]$

$$\frac{\partial^2 e_{22}}{\partial X_3 \partial X_1} = \frac{\partial}{\partial X_3} \cdot \frac{\partial e_{22}}{\partial X_1} = \frac{\partial}{\partial X_3} \cdot \frac{\partial}{\partial X_1}\left(2bX_2\right) = \frac{\partial}{\partial X_3}\left(0\right) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_2}\left(-\frac{\partial e_{31}}{\partial X_2} + \frac{\partial e_{12}}{\partial X_3} + \frac{\partial e_{23}}{\partial X_1}\right) = \frac{\partial}{\partial X_2}\left(-0 + 0 + 0\right) = 0 \quad (2)$$

$$(1) = (2)$$

$$0 = 0 \checkmark$$

$$\frac{\partial^2 e_{33}}{\partial X_1 \partial X_2} = \frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_2}\left(3mX_3^2\right) = \frac{\partial}{\partial X_1}\left(0\right) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_3}\left(-\frac{\partial e_{12}}{\partial X_3} + \frac{\partial e_{23}}{\partial X_1} + \frac{\partial e_{31}}{\partial X_2}\right) = \frac{\partial}{\partial X_3}\left(0 + 0 + 0\right) = 0 \quad (2)$$

$$(1) = (2)$$

$$0 = 0 \checkmark$$

$$2 \cdot \frac{\partial}{\partial X_1} \cdot \frac{\partial e_{33}}{\partial X_2} = 2 \cdot \frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_2}\left(3mX_3^2\right) = 2 \cdot \frac{\partial}{\partial X_1}\left(0\right) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_2} \cdot \frac{\partial}{\partial X_2}\left(aX_1 + bX_2\right) + \frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_1}\left(2bX_2\right) = \frac{\partial}{\partial X_2}\left(b\right) + \frac{\partial}{\partial X_1}\left(0\right) = 0 \quad (2)$$

$$(2) = (1) = 0 \checkmark$$

$$2 \frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_2}\left(\frac{1}{2}m\right) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_3} \cdot \frac{\partial}{\partial X_3}\left(2bX_2\right) + \frac{\partial}{\partial X_2} \cdot \frac{\partial}{\partial X_2}\left(3mX_3^2\right) = 0 + 0 = 0 \quad (2) \qquad (1) = (2) = 0 \checkmark$$

$$2 \cdot \frac{\partial}{\partial X_3} \cdot \frac{\partial}{\partial X_1}\left(0\right) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_1}\left(3mX_3^2\right) + \frac{\partial}{\partial X_3} \cdot \frac{\partial}{\partial X_3}\left(2bX_2\right) = 0 \quad (2) \qquad 1 = 2 = 0 \checkmark$$

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$U = a.r.\log(\theta) \quad V = a.r^2 + C.\operatorname{sen}(\theta) \quad W = 0$

$U_r = U.\cos\theta + V.\operatorname{sen}\theta$

$U_r = a.r.\log(\theta).\cos(\theta) + a.r^2.\operatorname{sen}(\theta) + C.\operatorname{sen}^2(\theta)$

$U_\theta = -U.\operatorname{sen}(\theta) + V.\cos(\theta)$

$U_\theta = -a.r.\log(\theta).\operatorname{sen}(\theta) + a.r^2.\cos(\theta) + C.\operatorname{sen}(\theta).\cos(\theta)$

El campo de desplazamiento dado por U, V y W es continuo y diferenciable

En la componente U, $\log(\theta)$ es discontinuo para $\theta = 0$

Se reduce el problema a $0 < \theta < 2\pi$ para evitar la discontinuidad y para que el campo sea valido. Como el campo es continuo y diferenciable es compatible

Considerando el campo de desplazamiento en coordenadas polares:

$$e_{rr} = \frac{\partial U_r}{\partial r} \qquad e_{\theta\theta} = \frac{U_r}{r} + \frac{1}{r}.\frac{\partial U_\theta}{\partial \theta}$$

$$e_{r\theta} = \frac{1}{2}.\left(\frac{1}{r}.\frac{\partial U_r}{\partial \theta} + \frac{\partial U_\theta}{\partial r} - \frac{U_\theta}{r}\right) \qquad e_{zr} = \frac{1}{2}\left(\frac{\partial U_r}{\partial z} + \frac{\partial U_z}{\partial r}\right)$$

$$e_{z\theta} = \frac{1}{2}\left(\frac{1}{r}\frac{\partial U_z}{\partial \theta} + \frac{\partial U_\theta}{\partial z}\right) \qquad e_{zz} = \frac{\partial U_z}{\partial z}$$

Donde $U(r,\theta) = \begin{bmatrix} U_r \\ U_\theta \\ U_z \end{bmatrix} = \begin{bmatrix} U(r,\theta) \\ V(r,\theta) \\ W(r,\theta) \end{bmatrix}$

Verificamos el campo que es compatible con la expresión S_{zz} en el plano

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a) Demostrar que el vector de rotación puede interpretarse físicamente como la rotación que sufre el entorno del punto considerado

Tenemos $\partial u_i = \frac{\partial u_i}{\partial x_j} \cdot \partial x_j$

$$\frac{1}{2}\frac{\partial u_i}{\partial x_j}\partial x_j + \frac{1}{2}\frac{\partial u_i}{\partial x_j}\partial x_j + \frac{1}{2}\frac{\partial u_j}{\partial x_i}\partial x_j - \frac{1}{2}\frac{\partial u_i}{\partial x_j}\partial x_j$$

$$= \underbrace{\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right)}_{\omega_{ij}}\partial x_j + \frac{1}{2}\left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j}\right)\partial x_j$$

$=$

$$= -\omega_{ij}\partial x_j + e_{ij}\partial x_j$$

$$= -\omega_{ij}\partial x_j$$

$$= -\varepsilon_{ijk}\,\omega_k\partial x_j$$

$$= \varepsilon_{ikj}\,\omega_k\partial x_j = (\omega \times \partial x)_i$$

b) $W_k = \frac{1}{2}\varepsilon_{kij} W_{ij} \longrightarrow W_{ij} = \varepsilon_{ijk} W_k$

$W_{ij} = \varepsilon_{ijk} W_k$ multiplico ambos lados

$\varepsilon_{ijp} W_{ij} = \varepsilon_{ijp}\varepsilon_{ijk} W_k$ Relación delta Epsilon

$\varepsilon_{ijp} W_{ij} = (\underbrace{\delta_{jk}\delta_{pj}} - \delta_{jj}\delta_{pk}) W_k$ $\delta_{jj} = 3$ y contraigo índices

$\varepsilon_{ijp} W_{ij} = (\delta_{kp} - 3\delta_{pk}) W_k$ distribuyo y cambio δ_{pk} a δ_{kp}

$\varepsilon_{ijp} W_{ij} = \underbrace{\delta_{kp} W_k} - 3\underbrace{\delta_{kp} W_k}$ contraigo índices

$\varepsilon_{ijp} W_{ij} = W_p - 3W_p$

$\varepsilon_{ijp} W_{ij} = -2W_p$

$-\frac{1}{2}\varepsilon_{ijp} W_{ij} = W_p$ roto índices y $W_{ij} = W_{ji}$

$-\frac{1}{2}\cdot -\varepsilon_{pji} W_{ji} = W_p$

$\frac{1}{2}\varepsilon_{pji} W_{ji} = W_p$ cambio índices para que quede igual que el enunciado

$\left[\frac{1}{2}\varepsilon_{kij} W_{ij} = W_k\right]$

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c) $\Omega_k = \varepsilon_{kij}\Omega_{ij} \longrightarrow \Omega_{ij} = \frac{1}{2}\varepsilon_{ijk}\Omega_k$

$\Omega_{ij} = \frac{1}{2}\varepsilon_{ijk}\Omega_k$ definición de vector de vorticidad

$= \frac{1}{2}\varepsilon_{ijk}\varepsilon_{k\ell m}\Omega_{\ell m}$

Reordenamos índices
Relación delta-Epsilon

$= \frac{1}{2}\varepsilon_{kij}\varepsilon_{k\ell m}\Omega_{\ell m}$

$= \frac{1}{2}\left(\delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}\right)\Omega_{\ell m}$ distributiva

$= \frac{1}{2}\left(\delta_{i\ell}\delta_{jm}\Omega_{\ell m} - \delta_{im}\delta_{j\ell}\Omega_{\ell m}\right)$ Contraemos índices

$= \frac{1}{2}\left(\Omega_{ij} - \Omega_{ji}\right)$ Por antisimetría del tensor de vorticidad

$= \frac{1}{2}\left(\Omega_{ij} + \Omega_{ij}\right)$

$= \frac{1}{2} \cdot 2\Omega_{ij}$

$= \Omega_{ij}$

d) Demostrar $\Omega = \mathrm{rot}(V)$

$$\mathrm{rot}(V) = \nabla \times V$$

$$\mathrm{rot}(V) = \varepsilon_{ijk} \frac{\partial V_k}{\partial X_j}$$

$$\mathrm{rot}(V) = \varepsilon_{ijk} \left(\frac{1}{2}\frac{\partial V_k}{\partial X_j} + \frac{1}{2}\frac{\partial V_k}{\partial X_j} + \frac{1}{2}\frac{\partial V_j}{\partial X_k} - \frac{1}{2}\frac{\partial V_j}{\partial X_k} \right)$$

$$\mathrm{rot}(V) = \varepsilon_{ijk} \frac{1}{2}\left(\frac{\partial V_k}{\partial X_j} - \frac{\partial V_j}{\partial X_k} \right) + \varepsilon_{ijk} \frac{1}{2}\left(\frac{\partial V_k}{\partial X_j} + \frac{\partial V_j}{\partial X_k} \right)$$ Reacomodo y multiplico definición Ω

$$\mathrm{rot}(V) = \varepsilon_{ijk} \Omega_k + \frac{1}{2}\left(\varepsilon_{ijk} \frac{\partial V_k}{\partial X_j} + \varepsilon_{ijk} \frac{\partial V_j}{\partial X_k} \right)$$ Cambio índices de ε_{ijk} a ε_{ikj}

$$\mathrm{rot}(V) = \varepsilon_{ijk} \Omega_k + \frac{1}{2}\left(\varepsilon_{ijk} \frac{\partial V_k}{\partial X_j} + \varepsilon_{ikj} \frac{\partial V_k}{\partial X_j} \right)$$

$$\mathrm{rot}(V) = \varepsilon_{ijk} \Omega_{jk} + \frac{1}{2}\left(\underbrace{\varepsilon_{ijk} \frac{\partial V_k}{\partial X_j}}_{1} - \underbrace{\varepsilon_{ijk} \frac{\partial V_k}{\partial X_j}}_{2} \right)$$

1 = 2 entonces

1 - 2 = 0

$$\mathrm{rot}(V) = \varepsilon_{ijk} \Omega_{jk} + \frac{1}{2} \cdot 0$$

$$\mathrm{rot}(V) = \varepsilon_{ijk} \Omega_{jk} \qquad [\Omega_k = \varepsilon_{kij} \Omega_{ij}]$$

$$\mathrm{rot}(V) = \Omega_i$$

$$\mathrm{rot}(V) = \Omega$$

e) $\partial V_i = \frac{\partial V_i}{\partial X_j} \partial X_j$

$$= \frac{1}{2} \frac{\partial V_i}{\partial X_j} \partial X_j + \frac{1}{2} \frac{\partial V_i}{\partial V_j} \partial X_j$$

$$= \frac{1}{2} \frac{\partial V_i}{\partial X_j} \partial X_j + \frac{1}{2} \frac{\partial V_i}{\partial X_j} \partial X_j + \frac{1}{2} \frac{\partial V_j}{\partial X_i} \partial X_j - \frac{1}{2} \frac{\partial V_j}{\partial X_i} \partial X_j$$

$$= \frac{1}{2} \left(\frac{\partial V_i}{\partial X_j} - \frac{\partial V_j}{\partial X_i} \right) \partial X_j + \frac{1}{2} \left(\frac{\partial V_j}{\partial X_i} + \frac{\partial V_i}{\partial X_j} \right) \partial X_j$$

$$= -\Omega_{ij} \partial X_j + V_{ij} \partial X_j$$

$$-\Omega_{ij} \partial X_j$$

$$-\frac{1}{2} \varepsilon_{ijk} \Omega_k \partial X_j$$

$$\frac{1}{2} \varepsilon_{ikj} \Omega_k \partial X_j$$

$$\frac{1}{2} (\Omega \times \partial x)_i$$